Wind Development and Residential Property Values: Historical Evidence and Future Implications

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Abstract

Wind power is central to the transition to low-carbon electricity, but concern about its effects on nearby property values is a common source of local opposition. We estimate how wind development has affected US home prices and use these estimates to assess the residential property value at stake as wind capacity expands. Using a stacked event study of home sales, we find that prices within 2 km of new commercial wind developments fell by about 11% relative to prices 5–10 km away in the year operation began and the following five years. We then place hypothetical turbines at the wind sites in four National Renewable Energy Laboratory scenarios for a clean US electricity system by 2035, which add about 600–1,200 GW of wind capacity. These scenarios would bring a turbine within 5 km of 2.5–6.3 million residential parcels that had none within 10 km in 2023, most of them in rural areas. If past price effects carry over to these homes, our estimates imply a reduction of $16–51 billion in their value across the scenarios, compared with $5–33 billion using estimates from two recent national studies. These findings show how the implications of wind expansion for property values depend on where turbines are built and on the size and reach of their price effects.

1 Introduction

The expansion of wind power is central to the transition to a low-carbon electricity system. The four scenarios developed by the National Renewable Energy Laboratory (NREL) for a clean US electricity system by 2035 add between about 600 and 1,200 GW of wind capacity to the 2020 fleet (Denholm et al., 2022). Development on this scale would affect not only the land on which turbines are built, but also the communities around them. Wind projects have increasingly been built closer to residential areas, where their effect on property values is a common concern and a source of opposition to further development (E. J. Brunner, Hoen, Rand, et al., 2024; Hoen, E. Brunner, et al., 2025).

This paper combines historical estimates of price changes around wind developments with a national assessment of the residential exposure implied by NREL's 2035 scenarios. We first estimate price changes around commercial wind developments using CoreLogic transaction records and a stacked event study (Cengiz et al., 2019; Wing et al., 2024). We compare homes within 2 km and 2–5 km of each development with homes 5–10 km away, before and after operation begins. Our main analysis covers years 0–5 of operation, and we use alternative samples, repeat sales, and proposals that were never built to help interpret the estimates.

We then place hypothetical turbines at NREL's sites according to their added capacity and use terrain-based viewsheds to measure potential visibility. For each residential parcel, we record distance, visibility, development size, and the number of other developments nearby. We value this exposure using our own estimates and those of two recent national studies, E. J. Brunner, Hoen, Rand, et al. (2024) and Guo et al. (2024), each of which retains its own proximity or visibility parameters. We adjust assessed values for differences in valuation practice using recent sale prices.

We find that prices fell close to wind developments relative to prices farther away. Our main estimate implies a relative price decline of about 11% within 2 km, and much of this

decline is already evident before operation begins, as in E. J. Brunner, Hoen, Rand, et al. (2024). The decline for rural homes is smaller, at about 6.2%, and changes less when we exclude the most heavily weighted developments. The pooled estimate is more sensitive to a few heavily weighted urban developments, and excluding the five with the largest regression weights reduces it to about 3.4%.

For the scenario-based future projections, we define previously unexposed parcels as those with no turbine within 10 km in 2023. The scenarios would place 2.5–6.3 million such parcels within 5 km of a new turbine. Most are rural, a majority would be near several developments, and the developments themselves are generally larger than those behind the historical estimates. Under the All Options scenario, applying our main estimates implies a conditional reduction of $35.4 billion in the value of these properties. The corresponding totals are $22.7 billion using the estimates of Brunner et al. and $9.6 billion using those of Guo et al. These differences reflect both the assumed percentage effect and the distances over which it applies.

Existing studies provide mixed evidence on how wind development affects property prices. Early studies in Cornwall and the United States found no statistically significant reductions in nearby property values (Hoen, Brown, et al., 2015; Hoen, Wiser, et al., 2011; Sims et al., 2008), while studies with larger samples close to turbines have more often found negative effects. Gibbons (2015) estimates a decline of 5–6% within 2 km of visible wind farms in England, and Jensen et al. (2014) find visibility and noise effects in Denmark. Estimates from the Netherlands, Germany, and Belgium vary with distance and setting (Dröes and Koster, 2016; Joly and De Jaeger, 2021; Sunak and Madlener, 2016). Negative effects have also been found in Sweden (Westlund and Wilhelmsson, 2025), and in Ontario, discounts were concentrated in municipalities that had opposed their projects (Vyn, 2018).

Recent US studies provide the closest comparisons. Dong et al. (2023) find declines of 2.5–4.6% within 1 km of turbines in Massachusetts and Rhode Island. E. J. Brunner, Hoen,

Rand, et al. (2024) compare distance bands within projects and estimate a decline of about 10.3% within one mile, averaged from announcement through ten years after announcement. Their event study shows the largest decline before operation, which falls to about 2% and becomes statistically insignificant roughly five years after operation begins. The effects are concentrated in counties within metropolitan areas of at least 250,000 people.

Guo et al. (2024) instead compare homes within 10 km of turbines that can and cannot see them. Their average visibility effect is -1.1%, reaching about -8% within 1.5 km. The decline peaks about three years after installation, diminishes over the following seven years, and is smaller for later installation cohorts. In contrast, Hoen, E. Brunner, et al. (2025) find that property values increase across school districts following wind development, excluding homes within 3.2 km in their preferred specification. Table 9 in appendix A summarizes these studies. Differences in their comparison groups, periods, and turbine populations help explain why they do not produce a single transferable estimate. Proximity effects can include changes before operation, visibility effects compare homes in the same local area, and district-wide gains may be shared by homes both near to and far from turbines.

This paper makes two main contributions. First, we measure the residential exposure implied by alternative wind deployment scenarios, including visibility and proximity to multiple developments, and compare that exposure with the homes behind the historical estimates. Second, we show how the resulting changes in property values depend on the empirical evidence used to value exposure. Even temporary price reductions affect owners who sell before values recover, which makes the location and persistence of these effects relevant to siting and compensation policy.

2 Theory

Housing prices reflect the value of a property's structural characteristics and the amenities associated with its location (Rosen, 1974). A wind development can affect these prices through the physical effects of its turbines, buyers' expectations about the project, and the local economic benefits it brings. Visual intrusion, noise, and shadow flicker arise once turbines are built, are strongest close to them, and may continue for as long as they operate (Jensen et al., 2014; Pedersen and Waye, 2004). Expectations can affect prices earlier, because a proposal may change buyers' assessment of an area before its consequences are known. This uncertainty can fade once the project is built or abandoned, and E. J. Brunner, Hoen, Rand, et al. (2024) consider this explanation for the decline before operation and subsequent recovery in their estimates. Lease payments, tax revenues, and improvements in public services can also be capitalized into property values (E. J. Brunner, Hoen, and Hyman, 2022; Hoen, E. Brunner, et al., 2025). Unlike the physical effects, these benefits do not fall mainly on nearby homes, since lease payments go to the landowners who host turbines and tax revenues are shared across a county or school district.

Prices may also recover after operation begins as households with different preferences move, with buyers who are less concerned about turbines replacing those who are more concerned (Banzhaf and Walsh, 2008; Tiebout, 1956). Preferences toward turbines vary, and Firestone and Kirk (2019) report that around 90% of Americans living within 8 km of a turbine would prefer it to a conventional power plant at the same distance.

These channels differ in when and where they affect prices. Expectations can move prices from the time a project is proposed, including around proposals that are never built, whereas the physical effects of turbines begin once they are built and are concentrated close to them. To see what a comparison of nearby and more distant homes measures, write the change in

the log price of home i associated with a development as

$$\Delta \ln p_i = a(d_i, v_i) + b_j, \tag{1}$$

where a depends on the home's distance d_i from the nearest turbine and on whether a turbine is visible (v_i), and b_j is an effect shared by all homes in the surrounding area j, such as that of higher local tax revenues. Comparing homes within 2 km of a development with homes 5–10 km away in the same area removes b_j and measures the difference in a between the two groups. This difference equals the effect close to the development when the turbines do not affect homes 5–10 km away, and it can be negative even where wind development raises property values across the wider area, as Hoen, E. Brunner, et al. (2025) find.

Capitalization of this kind need not equal households' willingness to pay for the change in local amenities, because prices also reflect sorting and other adjustments in the housing market (Kuminoff and Pope, 2014). It nevertheless measures the change in the value of the homes themselves, which is what owners gain or lose when they sell. Our projections aggregate this change across the homes that future developments would newly expose.

3 Data

Our analysis draws on property transactions and assessments, records of existing and proposed wind turbines, and the turbine sites in NREL's 2035 deployment scenarios. We first describe the property data and the geographic measures we attach to each property, and then the turbine data used in the historical analysis and the future deployment scenarios.

3.1 Property and Geographic Data

We use property data from CoreLogic (2024). These data include deed records, which report the price and date of each sale, and the history of assessment records, which report each property's location, type, and structural characteristics. Our sales panel covers 1970–2024 in the states with recorded turbines. We keep arm's-length, open-market sales and drop partial-interest transfers, leases, inter-family transfers, foreclosure and REO sales, short sales, sales below $1,000, and multi-parcel or split-parcel sales whose price cannot be assigned to a single parcel. We link each sale to the assessment records using CoreLogic's persistent parcel identifier (CLIP). Coordinates come from the parcel's most recent geocoded assessment, and bedrooms, bathrooms, and living area come from the first complete assessment in or after the year of sale.[1] We drop sales without a CLIP or a matching geocoded assessment, leaving a panel of 154.6 million sales.

We restrict our estimation samples to single-family homes, townhouses, condominiums, and multi-family buildings.[2] We also drop sales in the top and bottom 1% of the price distribution for each state and year. Our hedonic models require complete characteristics, which are available for 99.4 million sales (64% of the panel), while our repeat-sales models

[1] Where no complete assessment exists in or after the year of sale, we use the most recent earlier one. Characteristics taken from a later assessment may reflect changes made after the sale.

[2] Residential status is taken from the property-type code on the assessment in force at the time of sale or, where the code is missing, from the presence of complete housing characteristics on that assessment.

require only price and location. Maryland's assessment records report no bedroom counts, so Maryland enters only the repeat-sales analysis.

For the projections, we use the 2023 assessment snapshot, which covers every residential parcel with a recorded value and coordinates, including separate parcels that share a geocoded point. We value each parcel at its 2023 assessed value, CoreLogic's calculated total value. Assessed values are not a consistent measure of market value across the country, because jurisdictions assess at different fractions of market value, update their assessments at different intervals, and in some cases cap them. To put them on a common footing, we compare each parcel's 2023 assessed value with the price of its latest arm's-length residential sale in 2021–2023, adjusted to 2023 levels using the FHFA house price index (U.S. Federal Housing Finance Agency, 2024). Across these sales, the median ratio of assessed value to price is 0.80, but the ratio varies substantially across states (appendix D).

We divide each parcel's assessed value by the median ratio in its county, or by the median ratio in its state when the county has fewer than 30 usable sales or the parcel has no county code. We call the results assessment-ratio-adjusted values, or adjusted values for short. They correct for differences in assessment levels between areas but not within them, so we also report our projections using unadjusted assessed values.

We supplement the property data with measures of visibility and residential setting. We calculate visibility using the 1/3 arc-second elevation model from the USGS 3D Elevation Program, projected to 10 m cells (U.S. Geological Survey, 2024a). A home can see a turbine if there is a line of sight from 1.6 m above the ground at the home to the turbine's tip, allowing for earth curvature and atmospheric refraction. We evaluate visibility within 10 km of existing turbines and within 5 km of future turbines.[3] We assign each property to its

[3]Where a turbine's tip height is missing, we use its hub height plus half its rotor diameter and, failing that, the median tip height. The elevation model represents bare earth, so the viewsheds do not account for screening by buildings or vegetation. Where the elevation model has no value, we treat visibility as unresolved rather than as no view.

2020 census tract (U.S. Census Bureau, 2021) and classify tracts with at least 1,000 people per square mile as urban and the remainder as rural. Our township fixed effects use the six-by-six-mile townships of the Public Land Survey System where they exist and synthetic townships on the same grid elsewhere (appendix B).

3.2 Wind Developments

We use wind turbine data from the US Wind Turbine Database (U.S. Geological Survey, 2024b). These data include the location, commissioning year, capacity, and dimensions of approximately 75,000 turbines in the contiguous United States. We group turbines into *developments*, defined as turbines commissioned in the same year that are linked by distances of no more than 5 km, either directly or through intermediate turbines. Developments are defined geographically rather than by project name, so phases commissioned in different years form separate developments.

Our main sample includes only developments in which every turbine has a reported capacity above 600 kW. This threshold follows E. J. Brunner, Hoen, Rand, et al. (2024), but we apply it to whole developments rather than to individual turbines. We drop developments that contain any turbine reported at or below 600 kW, as well as *unclassified* developments, in which no turbine is known to be small but at least one is missing its capacity. Section 4.1 reports the sales lost under each rule. Turbines of every capacity still count toward each home's exposure history, so a home first exposed to a small or unclassified turbine is not treated as unexposed when a commercial development arrives later.

We identify developments that were proposed but never built using obstruction-evaluation filings from the Federal Aviation Administration (2025), which record proposed structures whether or not they are built. We use wind turbine filings from 2008–2016 for structures at least 250 feet tall and group them into proposed developments using the same rule as for built

turbines. We classify a proposal as never built if none of its filings is marked as built and no turbine in the US Wind Turbine Database lies within 5 km of any of its filings, which yields 214 never-built developments. We assign each a hypothetical operating year equal to its filing year plus the median lag between filing and operation among built projects.[4] Applying the event-study design to homes near these proposals provides a comparison with places where wind development was proposed but no turbines followed (in the spirit of Jarvis, 2025).

3.3 Future Deployment Scenarios

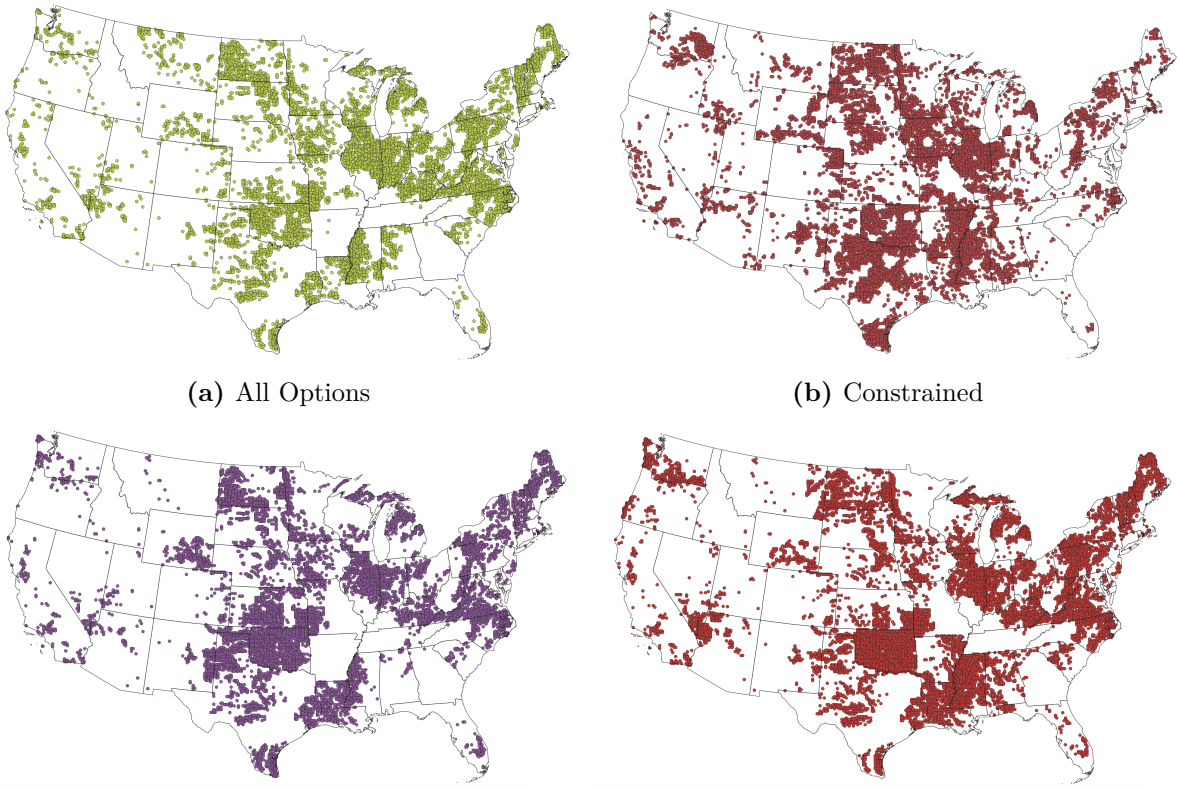
The future turbine sites come from NREL's ReEDS model runs for the four 2035 scenarios in Denholm et al. (2022), which were provided by Dr. Paul Denholm (National Renewable Energy Laboratory, 2024). These data include the location of each wind site and its capacity in 2020 and 2035. We model the capacity added between those years, which ranges from 614 GW in the Constrained scenario to 1,168 GW in the No CCS scenario. In the All Options scenario, all technologies improve according to NREL's 2021 Annual Technology Baseline, including direct air capture (National Renewable Energy Laboratory, 2021). The Infrastructure Renaissance scenario assumes improvements in transmission technology, permitting, and siting, while the Constrained scenario imposes additional limits on new generation and transmission. In the No CCS scenario, carbon capture and storage never becomes cost-competitive, making it the only scenario with zero direct power-sector emissions in 2035.

We represent each site's added capacity with hypothetical turbines that have a capacity of 3.1 MW and a tip height of 177 m, the medians for turbines installed in the database's three most recent years. Each site receives its added capacity divided by 3.1 MW, rounded to the nearest turbine. We place these turbines on a hexagonal grid, filled outward from the center of the site, at a density of 20 MW/km^2 (Enevoldsen and Jacobson, 2021), which corresponds to one turbine per 0.155 km^2 and a spacing of approximately 423 m. The scenarios contain

[4]The lag is one year for most filing years.

between 198,218 and 376,685 new turbines. Across all four scenarios, these occupy 574,573 distinct positions at 17,209 sites, and the largest development contains 127 turbines.[5] Figure 1 shows the sites in each scenario. Turbines installed by 2035 may be taller than this design, in which case our calculations may understate visibility.

Figure 1: Wind sites in NREL's four 2035 scenarios



(a) All Options

(b) Constrained

(c) Infrastructure

(d) No CCS

[5]The scenarios share turbine positions, so we compute each position's viewshed only once.

4 Historical Evidence

4.1 Empirical Strategy

We estimate price changes around each home's first wind development using a stacked event study. We estimate separate models that compare homes within 2 km and 2–5 km of a development's nearest turbine with homes 5–10 km away. In both groups, the development must be the first to bring a turbine within 10 km of the home, which prevents previously exposed homes from serving as comparisons for newly exposed homes (Cengiz et al., 2019).

Event time $e = t - c_s$ is the year of sale t minus the development's first operating year c_s. The main window runs from year -6 to year 5, so the five-year follow-up covers the six operating years 0–5. Years -6 to -3 form the baseline, and we estimate years -2 and -1 separately as the development phase. For each distance band, we estimate

$$\ln p_{ist} = \sum_{e \neq -3} \beta_e \, \mathbb{1}[t - c_s = e] \, D_{is} + X_{it}'\gamma + \alpha_{s \times t} + \alpha_{s \times \tau(i)} + \alpha_{s \times D} + \lambda_q + \varepsilon_{ist}, \tag{2}$$

where p_{ist} is the price of home i sold in year t in the stack for development s, and D_{is} identifies treated homes. The controls X_{it} contain the logs of floor area, bathroom count, bedroom count, and lot size, together with age, age squared, and indicators for missing lot size and age. We include fixed effects for development-by-year, development-by-township, development-by-treatment-group, and season of sale, where $\tau(i)$ identifies the township and q the season of the transaction. We use the corrective weights of Wing et al. (2024) and cluster standard errors by county. The resulting coefficients are regression-weighted contrasts, which need not equal averages of the development effects weighted exactly by their shares of treated sales, and appendix C reports the implicit weights.

Our main estimate is the mean for years 0–5 less the mean for the baseline years:

$$\theta_{\text{early}} = \frac{1}{6}\sum_{e=0}^{5}\beta_e \;-\; \frac{1}{4}\sum_{e=-6}^{-3}\beta_e, \qquad \beta_{-3} \equiv 0. \tag{3}$$

A causal interpretation requires that, conditional on the controls and fixed effects, prices in the treated and comparison bands would have followed parallel paths in the absence of development, and that the baseline precedes any differential response to the proposal. Due to the fact that the comparison removes effects shared equally by near and far homes, such as a community-wide financial gain, the estimates capture the additional price response close to a development. The pre-development coefficients, alternative samples, and never-built proposals provide evidence on these assumptions.

We exclude homes with an undated turbine within 10 km, homes first exposed to several developments in the same year, and developments more than 30 km wide. We also exclude homes that a later development brings into a closer distance band before the window ends. Counties must have deed coverage in every year of the window, which we define as at least 20% of the county's median annual sales.

We focus on commercial developments because their turbines are more comparable to future utility-scale development. The five-year follow-up admits developments through 2019, compared with 2014 under a ten-year requirement, while retaining the observations needed for the years 0–5 contrast. We also report results for developments of every size and for a ten-year window.

Our main sample within 2 km contains 34,366 treated sales around 172 commercial developments commissioned between 2000 and 2019 (table 1). Developments commissioned from 2013 account for 19% of treated sales, and the 50 developments commissioned from 2015, which could not enter a ten-year window, contribute 5,141 treated sales (15%).

Most treated sales are near small developments. Single-turbine sites account for 65% of

treated sales and developments with ten or more turbines for 14%, even though the median commercial development with treated sales in either distance band contains 21 turbines. This difference reflects the greater number of homes near single-turbine sites. The median turbine capacity is 1.5 MW, and California, Massachusetts, and Ohio account for 23%, 22%, and 16% of treated sales, respectively.

The sample of developments of every size contains 253 developments and 108,916 treated sales, 84% of which are near single turbines, and its median turbine capacity is 0.5 MW. Table 2 separates the exclusions that produce the commercial sample. Of the 74,550 excluded treated sales, 66,436 are near developments with a turbine reported at or below 600 kW, and 8,114 are near developments excluded only because a capacity is missing. Both groups consist mainly of single-turbine developments, but a missing capacity alone does not make a development noncommercial.

Table 1: Historical estimation samples (homes within 2 km)

	Commercial, five-year (main)	Every size, five-year	Commercial, ten-year	Every size, ten-year	Never-built proposals, five-year
Developments	172	253	125	189	54
Treated sales	34,366	108,916	41,181	140,244	14,205
Operating from	2000–2019	2000–2019	2000–2014	2000–2014	2010–2016
Single turbine (%)	65	84	69	86	71
10+ turbines (%)	14	4	13	4	19
Median turbine (MW)	1.50	0.50	1.50	0.23	
Small / unclassified (%)	0 / 0	61 / 7	0 / 0	64 / 6	
From 2013 (%)	19	12	5	3	5
Leading states	CA 23% MA 22% OH 16%	MA 32% OH 17% MN 8%	MA 26% CA 18% OH 16%	MA 35% OH 17% MN 10%	RI 19% CT 19% IA 16%

Notes: For each specification, the table describes the sample within 2 km, including the developments with treated sales, their treated sales and commissioning years, the shares of sales by development size, median turbine capacity, and leading states. The small and unclassified shares refer to the capacity classes in table 2 and are zero in the commercial samples. The share from 2013 covers developments commissioned in or after 2013.

Table 2: Development capacity classes, five-year window

	Commercial (main)	Excluded: known small turbine	Excluded: capacity missing	All developments
Developments in the stacks	577	94	53	724
Developments with treated sales				
With treated sales	251	68	34	353
Median turbine count	21	1	1	3
One turbine	53	62	27	142
2–9 turbines	49	5	5	59
10–49 turbines	56	1	1	58
50+ turbines	93	0	1	94
Treated sales within 2 km	34,366	66,436	8,114	108,916
Treated sales at 2–5 km	178,877	301,549	61,156	541,582

Notes: Commercial developments have a reported capacity above 600 kW for every turbine. Developments in the second column have at least one turbine reported at or below 600 kW, and those in the third have a missing capacity but no known small turbine. Stack counts refer to developments forming five-year stacks. Counts of treated sales and development sizes use the samples of developments of every size in each distance band, and pruning after the commercial restriction can change the main samples slightly.

4.2 Results

Prices within 2 km of new developments fell by about 11% relative to prices 5–10 km away between the baseline and years 0–5. At 2–5 km, the coefficient implies a smaller decline of about 1.9%, but its interval runs from a decline of about 6% to an increase of about 2%. Table 3 reports the phase averages, and figure 2 shows the annual coefficients.

Much of the decline within 2 km is already present by year −3. Table 4 expresses each phase relative to that reference year. The baseline mean coefficient, including year −3 at zero, is 0.083, while the means for the development phase and years 0–5 are −0.079 and −0.033, respectively. Years −6 to −4 alone have a mean coefficient of 0.110. Our estimate of −0.116 therefore subtracts the baseline mean of 0.083 from the years 0–5 mean of −0.033.

These earlier movements could reflect anticipation of development, differences in local

housing market trends, or sampling noise. Prices may respond to a proposal before operation, as in E. J. Brunner, Hoen, Rand, et al. (2024), but the early coefficients are imprecise, with standard errors of 0.07–0.09 compared with 0.03–0.04 after operation. The joint pre-trend test does not reject ($p = 0.47$), but the intervals are wide enough to include sizable pre-development movements.

Price paths differ between rural and urban homes. For rural homes, the baseline coefficient is close to zero, and prices fall during development and remain lower in years 0–5. The rural estimate implies a decline of about 6.2%. The urban coefficient of -0.175 corresponds to a decline of about 16.1%, but its baseline mean coefficient is 0.172 and its mean in years 0–5 is only -0.003. We do not formally test the difference between the rural and urban estimates.

The urban sample contains 19 developments, 18,806 treated sales, and 20 county clusters, and its regression weights are concentrated, with one 750 kW turbine in Chicago carrying 22% of the weight on years 0–5. The pooled estimate is sensitive to a few heavily weighted developments (appendix table 11). Removing the five with the largest weights reduces the estimated decline from about 11% to 3.4%. Four of these developments are single turbines and one has three turbines, and all five are in metropolitan counties. Together they carry 43% of the pooled model's weight on years 0–5 and 78% of the urban model's. Removing the Chicago turbine alone gives a coefficient of -0.066. In contrast, excluding the developments assigned to California or Massachusetts gives larger negative coefficients of -0.156 and -0.139, respectively. The refit without the five developments is also more precise than the main model.

The rural estimate is more stable across geographical exclusions. Removing its five most heavily weighted developments gives a coefficient of -0.058. Across nine refits, which exclude each of these five developments, all five together, and the developments assigned to each of three states, the coefficient ranges between -0.077 and -0.056, and every interval excludes zero.

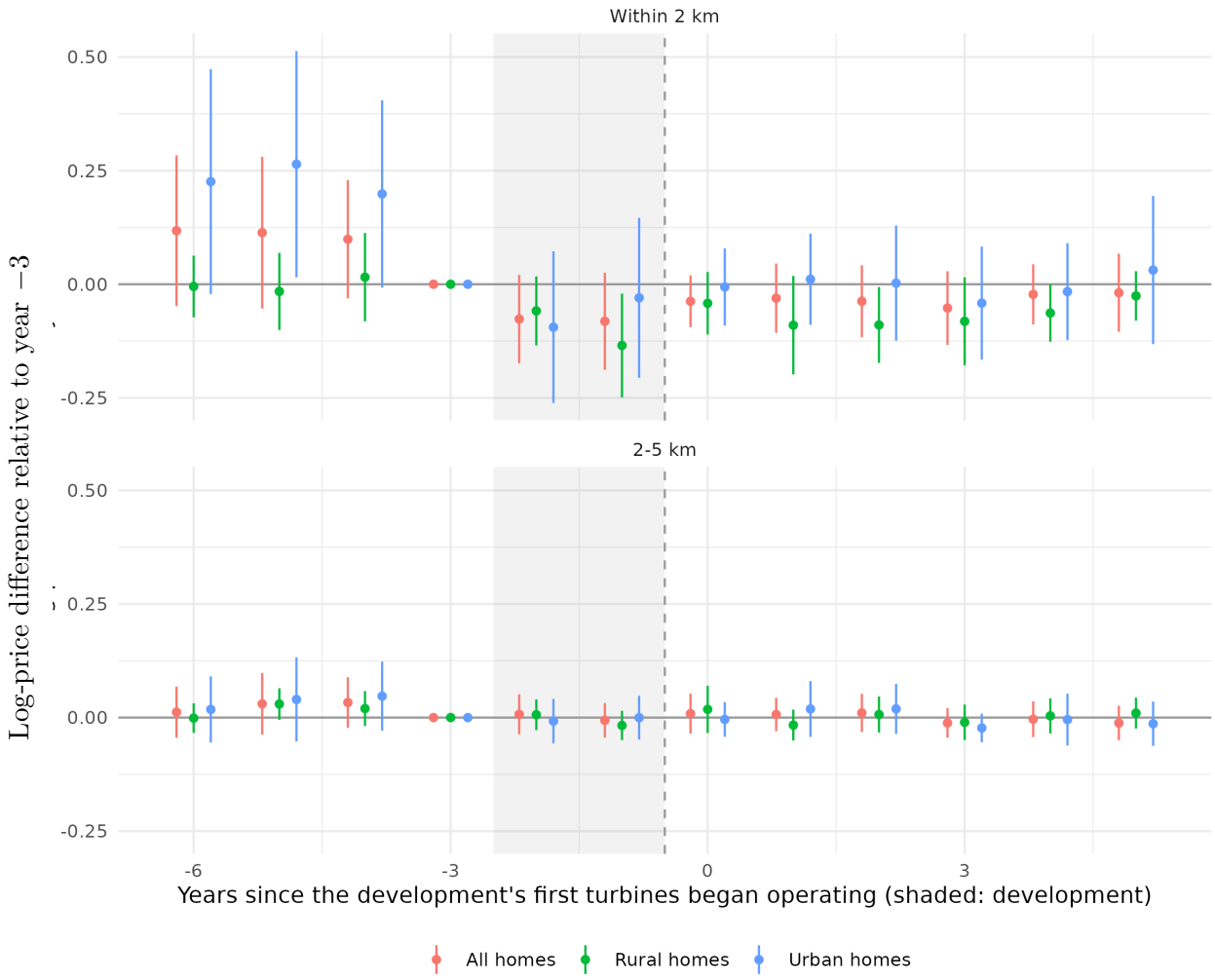
Table 3: Estimated price changes near new wind developments, relative to the pre-development baseline

	Within 2 km	2–5 km	Pre-trend p	Developments	Treated sales
A. Main estimate: commercial developments, five-year window					
All homes	-0.116^{**}	-0.019	0.47	172	34,366
	(0.055)	(0.022)			
Rural homes	-0.064^{***}	-0.010	0.71	164	14,392
	(0.021)	(0.013)			
Urban homes	-0.175^{*}	-0.027	0.11	19	18,806
	(0.084)	(0.031)			
B. Comparisons					
Developments of every size and class, five-year window	-0.061^{**}	-0.008	0.51	253	108,916
	(0.026)	(0.011)			
Commercial, ten-year window: years 0–5	-0.121^{**}	-0.021	0.32	125	41,181
	(0.061)	(0.025)			
years 6–10	-0.043	-0.014			
	(0.037)	(0.014)			
Every size, ten-year window: years 0–5	-0.064^{**}	-0.015	0.49	189	140,244
	(0.029)	(0.014)			
years 6–10	-0.002	0.010			
	(0.025)	(0.012)			
C. Checks					
Repeat sales	-0.142^{**}	-0.048	0.40	183	21,896
	(0.072)	(0.052)			
Proposals never built	-0.058	-0.035	0.22	54	14,205
	(0.046)	(0.025)			
Turbine in view (within 2 km; nearest in view 2–5 km)	-0.116^{**}	-0.020	0.50	172	33,690
	(0.056)	(0.023)			

Notes: Each entry is the change in log sale price for homes within 2 km or 2–5 km of a development, relative to homes 5–10 km from the same development, between the baseline years −6 to −3 and years 0–5 after operation begins (or years 6–10 where stated). Table 4 shows how the estimates within 2 km are formed. County-clustered standard errors are in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The pre-trend test compares years −6 to −4 jointly with year −3. Counts and pre-trend tests refer to the model within 2 km. Models are hedonic unless stated otherwise.

Including developments of every size reduces the estimated decline within 2 km to about 5.9%. The coefficient is −0.061 within 2 km and −0.008 at 2–5 km. In this sample, the baseline mean coefficient is close to zero, followed by a mean of −0.062 in years 0–5 (figure 3).

Figure 2: Property prices around new commercial wind developments



Notes: Five-year follow-up. The panels show homes within 2 km and at 2–5 km, and the series distinguish all, rural, and urban homes. Points show annual log-price coefficients relative to year −3, with 95% confidence intervals. Comparison homes are 5–10 km from the same development, and shaded years are the development phase.

Table 4: Decomposition of the estimates within 2 km into period averages relative to year −3

	Relative to year −3				Reported estimate
	Baseline (−6 to −3)	Development (−2, −1)	Years 0–5	Years 6–10	(years 0–5 minus baseline)
Main estimate: all homes	0.083 [−0.031, 0.197]	−0.079 [−0.177, 0.019]	−0.033 [−0.094, 0.028]		−0.116 [−0.224, −0.007]
Rural homes	−0.001 [−0.045, 0.042]	−0.097 [−0.183, −0.010]	−0.065 [−0.127, −0.004]		−0.064 [−0.105, −0.023]
Urban homes	0.172 [−0.002, 0.346]	−0.062 [−0.231, 0.107]	−0.003 [−0.101, 0.094]		−0.175 [−0.352, 0.001]
Every size and class	−0.001 [−0.047, 0.045]	−0.041 [−0.088, 0.005]	−0.062 [−0.115, −0.009]		−0.061 [−0.111, −0.011]
Repeat sales	0.109 [−0.022, 0.241]	−0.087 [−0.198, 0.025]	−0.032 [−0.092, 0.027]		−0.142 [−0.283, −0.000]
Proposals never built	0.014 [−0.075, 0.103]	−0.032 [−0.086, 0.022]	−0.044 [−0.082, −0.006]		−0.058 [−0.150, 0.034]
Commercial, ten-year window	0.094 [−0.029, 0.218]	−0.083 [−0.199, 0.032]	−0.027 [−0.102, 0.049]	0.052 [−0.070, 0.173]	−0.121 [−0.241, −0.001]

Notes: Each row shows how an estimate within 2 km in table 3 is formed. The first four columns give the average annual coefficient in each period, measured against year −3, which enters the baseline mean at zero. The final column is the estimate reported in table 3, the years 0–5 mean minus the baseline mean. Brackets show 95% intervals based on the county-clustered covariance and degrees of freedom.

The two samples differ in their estimated price paths as well as in turbine size and location.

The ten-year window suggests a smaller decline in later years, but cannot distinguish persistence from full recovery (appendix figure 5). Among commercial developments commissioned by 2014, the estimates imply declines of about 11.4% in years 0–5 and 4.2% in years 6–10. The mean coefficient for years 6–10 relative to year −3 is 0.052, compared with a baseline mean of 0.094, so recovery must be judged against the baseline. For developments of every size, the corresponding coefficients are −0.064 and −0.002. The intervals are wide enough to allow a loss several years after operation. Comparing early and late estimates within this older population holds the set of developments fixed, whereas comparing its late estimate with our main estimate from the five-year window would also reflect differences between the two sets of developments.

Repeat-sales models, which replace the hedonic controls with property-by-development fixed effects, compare transactions for the same property. The estimate within 2 km implies a relative decline of about 13.2%, and the coefficient at 2–5 km is −0.048 (appendix figure 7). Within 2 km, the baseline mean coefficient is 0.109 and the years 0–5 mean is −0.032, both relative to year −3, so controlling for time-invariant property characteristics does not remove the main estimate's dependence on earlier price differences.

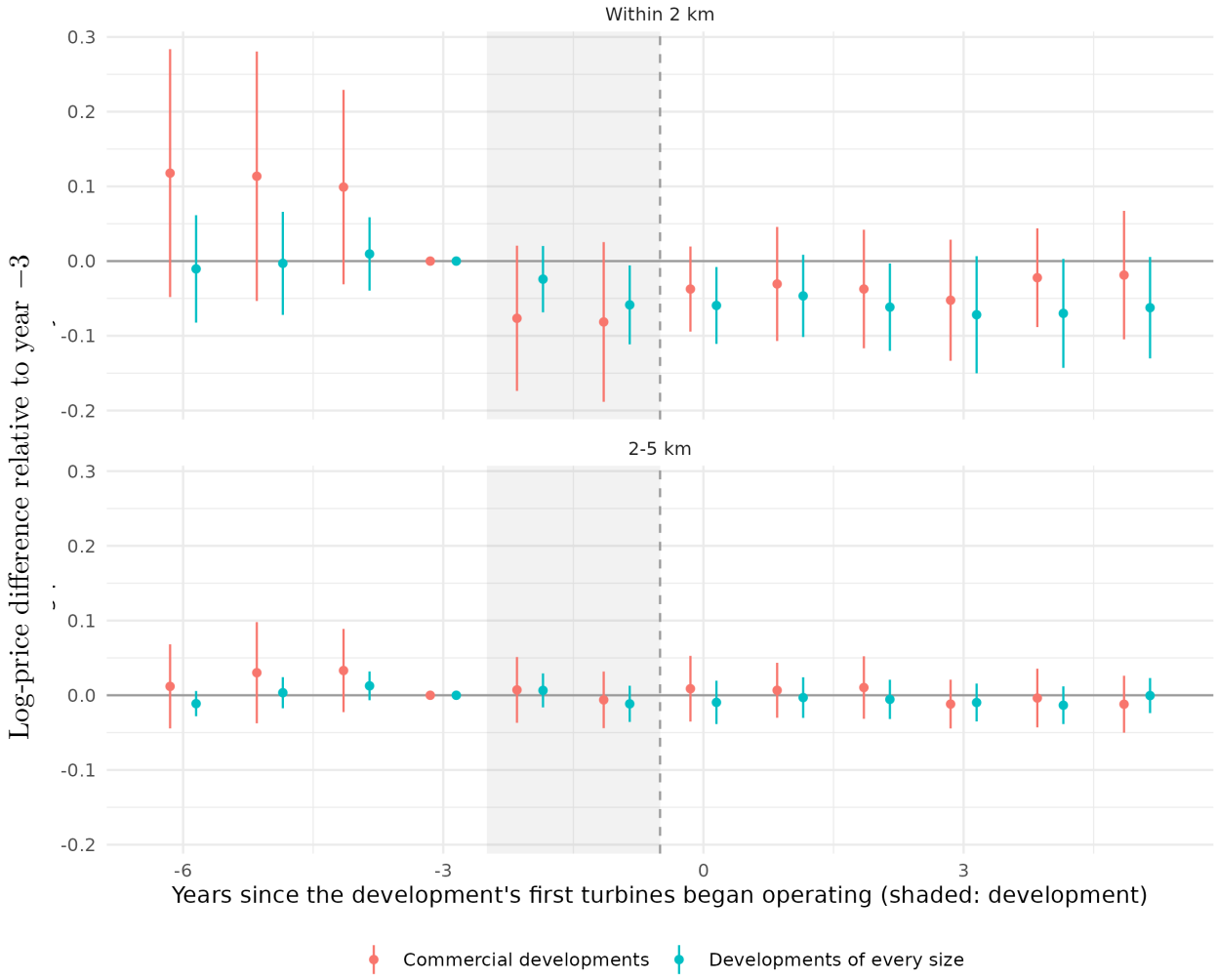
Prices also decline around never-built proposals, although the estimates are imprecise. For the 54 proposals with sufficient sales within 2 km, the estimate for hypothetical operating years 0–5 implies a decline of about 5.6%, and the coefficient at 2–5 km is −0.035. Their baseline mean coefficient is 0.014 and their mean in years 0–5 is −0.044, compared with −0.033 around built developments (appendix figure 6). These paths are too imprecisely estimated to tell apart.

Never-built proposals can still affect expectations, generate controversy, and create uncertainty. Their price paths therefore limit how much of the decline we can attribute to operating turbines, while leaving open an effect of the development process itself.

Visibility adds little within 2 km, where homes with a visible turbine account for 98% of the distance model's treated sales and give a coefficient of −0.116. The coefficient for homes whose nearest visible turbine is 2–5 km away is −0.020. The estimates are too similar to separate a visual mechanism from proximity, and we use distance as our main exposure measure. An exploratory interaction with the log of the number of turbines is also imprecise. Within 2 km, the estimated years 0–5 contrast increases by 0.004 per doubling of development size, which leaves substantial uncertainty about effects around the larger developments in the scenarios.

The regression weights are concentrated across developments (appendix C). Their concentration is equivalent to about 20 equally weighted developments in the main model, 37 for rural homes, and 7 for urban homes. These are measures of concentration rather than counts of observations or independent clusters. The largest development accounts for 12% of the main model's weight on years 0–5, whereas no development would receive more than 9% under target weights based on each development's share of treated sales. At the development-year level, the distance between the regression weights and these target weights is 0.13, measured as half the summed absolute difference.

Figure 3: Commercial developments and developments of every size



Notes: Five-year follow-up. Points show annual log-price coefficients relative to year −3, with 95% confidence intervals. Comparison homes are 5–10 km from the same development, and shaded years are the development phase.

5 Future Exposure

5.1 Exposure Measurement

For each scenario, we measure exposure at every assessment location within 10 km of a hypothetical turbine, which covers 33.3 million locations across the scenarios. We assign each property to its nearest new turbine and that turbine's development, and we record its distance, the size of the development, its visibility, the number of developments within 10 km, and the distance to the nearest turbine operating in 2023. We count properties near several new developments once, rather than applying a separate full discount for each development.

A property has a view when at least one turbine of its assigned development is confirmed visible within 5 km. Where no view is confirmed but missing elevation data prevent a determination, we classify visibility as unresolved, which distinguishes uncertainty about visibility from an established lack of a view.

5.2 Results

The scenarios place 6 to 11.4 million property locations within 5 km of a new turbine, and 15.6 to 24.8 million within 10 km. Table 5 restricts these counts to residential parcels with recorded values. Consistent with the historical analysis of first exposure, our main population consists of previously unexposed parcels, which have no turbine within 10 km in 2023.

Between 2.5 million parcels in the Constrained scenario and 6.3 million in the No CCS scenario would be newly exposed within 5 km, and 0.45–1.30 million of these would be within 2 km. The assessed values of the newly exposed parcels within 5 km total $311–895 billion, or $450–1,387 billion in assessment-ratio-adjusted values. Between 88% and 92% of these parcels have a terrain-based line of sight to a turbine of their nearest development, and none has unresolved visibility. Only 13–18% are in urban tracts. A further 0.57–0.77 million parcels

within 5 km of a new turbine already have a turbine within 10 km, and we report these separately.

Exposure is geographically concentrated, but its location varies by scenario. Virginia, Illinois, Pennsylvania, North Carolina, and Indiana each contain 8–10% of newly exposed parcels in the All Options scenario. Texas accounts for 16% in the Constrained scenario, while Oklahoma accounts for 13% in the Infrastructure and No CCS scenarios. The implications for individual communities therefore depend on the deployment pathway as well as the national amount of wind capacity.

Table 5: Residential exposure to the 2035 scenarios' new turbines

	All Options	Constrained	Infrastructure	No CCS
Parcels within 2 km (thousands)	909	449	813	1,297
Parcels 2–5 km (thousands)	3,522	2,040	3,060	4,960
Value within 5 km, assessed ($ billion)	611.1	311.2	530.9	895.0
Value within 5 km, assessment-ratio-adjusted ($ billion)	970.3	450.2	833.7	1387.2
Assessed value adjusted by county / state / not adjusted (%)	97 / 3 / 0	95 / 5 / 0	96 / 4 / 0	97 / 3 / 0
Near several developments (%)	70	59	67	73
Turbine of the nearest development in view (%)	88	92	90	89
View unresolved (%)	0	0	0	0
Value with the view unresolved ($ billion)	0.0	0.0	0.0	0.0
In urban tracts (%)	15	13	18	17
Already within 10 km of a turbine: parcels (thousands)	683	567	705	766

Notes: Unless stated otherwise, residential parcels have no turbine within 10 km in 2023. Distances refer to the nearest new scenario turbine. Adjusted values use county or state assessment ratios (section 3.1), and adjustment shares use assessed value as the denominator. A view requires a confirmed visible turbine of the assigned development within 5 km. Unresolved visibility means that no view is confirmed and missing elevation data prevent a determination.

The future scenarios differ from the estimation sample in both development size and residential setting (table 6). Scenario sites average 21–40 turbines and contain as many as 127. Among newly exposed parcels within 2 km, almost none are near a single turbine

and 23–75% are near developments with at least ten turbines, compared with 65% and 14% of treated sales in the historical sample. Urban tracts account for 57% of treated sales in the separate rural and urban estimation samples, compared with 9–14% of newly exposed parcels within 2 km. The historical urban share uses these subgroup samples, while the other historical comparisons use the pooled sample.

Multiple nearby developments are also more common in the scenarios. Between 62% and 74% of property locations within 5 km of a new turbine have more than one new development within 10 km, as do 50–71% of newly exposed residential parcels within 2 km. These counts differ from the historical exclusion of homes first exposed to several developments in the same year, since a staggered rollout could expose a home to one development before the others. Nevertheless, they identify settings for which an estimate based on isolated first developments provides limited guidance.

Restricting the historical sample to commercial developments does not remove these differences in development size and residential setting, since the five developments carrying the most weight in the main estimate are still small sites in metropolitan counties. The rural sample is closer to the future setting, with 34% of treated sales near developments of ten or more turbines and 43% near single turbines, compared with zero and 84% in the urban sample. Applying any of these estimates nationally therefore requires assumptions about how effects vary between the settings represented in the data and those projected for the future.

Table 6: Historical and future exposure within 2 km

	Historical (main estimate)	All Options	Constrained	Infrastructure	No CCS
Treated sales / parcels within 2 km	34,366	909,460	448,655	812,615	1,297,120
Nearest development a single turbine (%)	65	0	2	0	0
Ten or more turbines (%)	14	69	23	75	69
Fifty or more turbines (%)	11	3	0	4	2
In urban tracts (%)	57	10	9	13	14
Several developments within 10 km (%)		68	50	67	71

Notes: Historical observations are treated sales within 2 km in the main model, except for the urban share, which is calculated from the separately estimated subgroup samples as 18,806 urban sales divided by 18,806 urban plus 14,392 rural sales. Their combined sample is smaller than the pooled sample of 34,366 because the subgroup models exclude homes without a census density and are pruned separately. Scenario observations are residential parcels with no turbine within 10 km in 2023 and a new turbine within 2 km. Development size refers to the development containing the nearest turbine.

6 Exposure Valuation

This section combines the historical estimates with the exposure measured in section 5. For each scenario, we ask how much the value of the newly exposed homes would fall if they experienced the price changes estimated for past wind developments. The answer depends on which estimates are applied, so we repeat the calculation with our own estimates and with those of two recent national studies.

6.1 Methodology

For each newly exposed home, we multiply its value by the proportional fall in price implied by a set of estimates, which we call a profile, and add up the results. The price change depends on the home's distance from the nearest new turbine or, for one profile, on whether it can see a turbine. For a given scenario and profile, the total reduction in value is

$$L = \sum_i V_i \left[1 - \exp(\Delta g_i)\right], \tag{4}$$

where the sum runs over the newly exposed homes, V_i is the adjusted value of home i, Δg_i is the change in log price that the profile assigns to it, and $1 - \exp(\Delta g_i)$ converts that change into a proportional fall in value. A positive total is therefore a reduction in value and a negative total an increase.

The calculation covers the homes that would be newly exposed, which have no turbine within 10 km in 2023 and a new turbine within 5 km. These correspond to the first exposures in the historical analysis, and we report homes that already have a turbine nearby separately. The total is a one-time change in the value of the exposed homes, which lasts as long as the price effect does. It is not an annual loss or a measure of welfare. Our estimates compare homes near a development with homes 5–10 km away, so the calculation leaves out changes

shared by both groups, such as the effect of higher local tax revenues discussed in section 2. NREL's scenarios give the location and capacity of each wind site in 2035 but not when it would be built, so the results refer to the years after each development begins operating rather than to a calendar year. The calculation assumes that the price effects estimated for past developments apply to homes near future ones.

6.2 Price Effects

Table 7 lists the price effects we apply. Our main profile uses the estimates from section 4, a change in log price of −0.116 for homes within 2 km and −0.019 for homes 2–5 km away, both averaged over years 0–5 after operation begins. To look beyond year 5, we also apply the estimates from the ten-year window, which come from developments that began operating by 2014 and cover years 0–5 and 6–10.

The Brunner profile applies the national estimates of E. J. Brunner, Hoen, Rand, et al. (2024), a change in log price of −0.109 within one mile (1.6 km) and −0.031 at one to two miles. Their estimate for two to three miles is positive but statistically insignificant (+0.014), so we set it to zero. Their estimates average over the period from a project's announcement to ten years after it. The Guo profile applies the average effect of a turbine view estimated by Guo et al. (2024), a change in log price of −0.011, to homes that can see a turbine of their nearest new development within 5 km. Guo et al. measure views of turbine hubs within 10 km, while we measure views of blade tips within 5 km, so this profile approximates their measure.

Neither published estimate refers specifically to years 0–5 or 6–10 after operation. For years 6–10, we therefore bracket the published averages with two assumptions: full recovery, under which the effect is zero, and no recovery, under which it stays at the published average. Two further profiles, which we added after examining the main results, explore residential

setting and distance. The first applies our separate rural and urban estimates in each distance band, because most newly exposed homes are rural. The second applies the main estimate within 2 km and assumes no effect beyond 2 km. A final benchmark assumes no effect at all.

Table 7: Price effects applied to newly exposed homes

Profile	Exposure	Period	Log-price change (SE)
Estimated in this paper			
Main estimate	Within 2 km	Years 0–5	−0.116 (0.055)
	2–5 km	Years 0–5	−0.019 (0.022)
Ten-year window	Within 2 km	Years 0–5; 6–10	−0.121 (0.061); −0.043 (0.037)
	2–5 km	Years 0–5; 6–10	−0.021 (0.025); −0.014 (0.014)
Published averages			
Brunner et al. (2024)	Within 1 mile (1.6 km)	From announcement through 10 years after announcement	−0.109 (0.039)
	1–2 miles	Same period	−0.031 (0.020)
Guo et al. (2024)	Turbine in view (within 5 km here)	After installation	−0.011 (0.004)
Persistence assumptions for the published averages			
Full recovery	As above	Years 6–10	0
No recovery	As above	Years 6–10	Published average
Exploratory profiles			
By area and band	Rural: within 2 km; 2–5 km	Years 0–5	−0.064 (0.021); −0.010 (0.013)
	Urban: within 2 km; 2–5 km	Years 0–5	−0.175 (0.084); −0.027 (0.031)
Within 2 km only	Within 2 km; beyond	Years 0–5	−0.116 (0.055); 0
No effect	All exposures	All periods	0

Notes: Effects are changes in log price, and years refer to time since operation unless stated otherwise. Our estimates are from table 3. The published estimates are from Brunner et al.'s Table 4, column 8, and Guo et al.'s Table 1, column 1. Brunner et al.'s positive estimate for 2–3 miles (+0.014) is set to zero. Guo et al. measure views of turbine hubs within 10 km, while we measure views of blade tips within 5 km. The rural and urban profile classifies tracts with at least 1,000 people per square mile as urban and treats tracts with missing density as rural. Persistence and the absence of effects beyond 2 km are assumptions.

The intervals we report reflect sampling uncertainty in these price effects. We draw the coefficients 10,000 times from a joint normal distribution and recalculate each total,

holding home values and scenario layouts fixed (Krinsky and Robb, 1986). For our estimates, the draws use the county-clustered covariance of the two distance-band estimates from a combined regression (their correlation is 0.84), and the rural and urban profile uses the joint covariance of all four models. E. J. Brunner, Hoen, Rand, et al. (2024) do not report the covariance between their estimates for different distance bands, so we show intervals under both independence and perfect positive correlation.

6.3 Results

Table 8 and figure 4 report the results. Under our main estimates, the newly exposed homes in the All Options scenario would lose $35.4 billion, or 3.7% of their adjusted value. The total rises with the amount of exposure in each scenario, from $15.6 billion in the Constrained scenario to $51.0 billion in the No CCS scenario. These totals are imprecise, and their intervals include zero in every scenario. In All Options, the interval runs from a $16.5 billion increase to an $82.7 billion reduction in value. The estimate within 2 km excludes zero, but the estimate for 2–5 km, which applies to about four fifths of the exposed value, allows for gains as well as losses.

The published estimates give smaller totals, $22.7 billion with the Brunner profile and $9.6 billion with the Guo profile in All Options. The totals differ because the profiles differ in both the size of the effect and the distances over which it applies. Under our main estimates, homes 2–5 km away contribute $14.8 billion, or 42% of the total, despite their smaller percentage decline, because there are many more of them. The Brunner profile assigns no effect beyond two miles (3.2 km), while the Guo profile applies the same small effect to every home with a view, so homes 2–5 km away account for $7.6 billion, or 79%, of the Guo total.

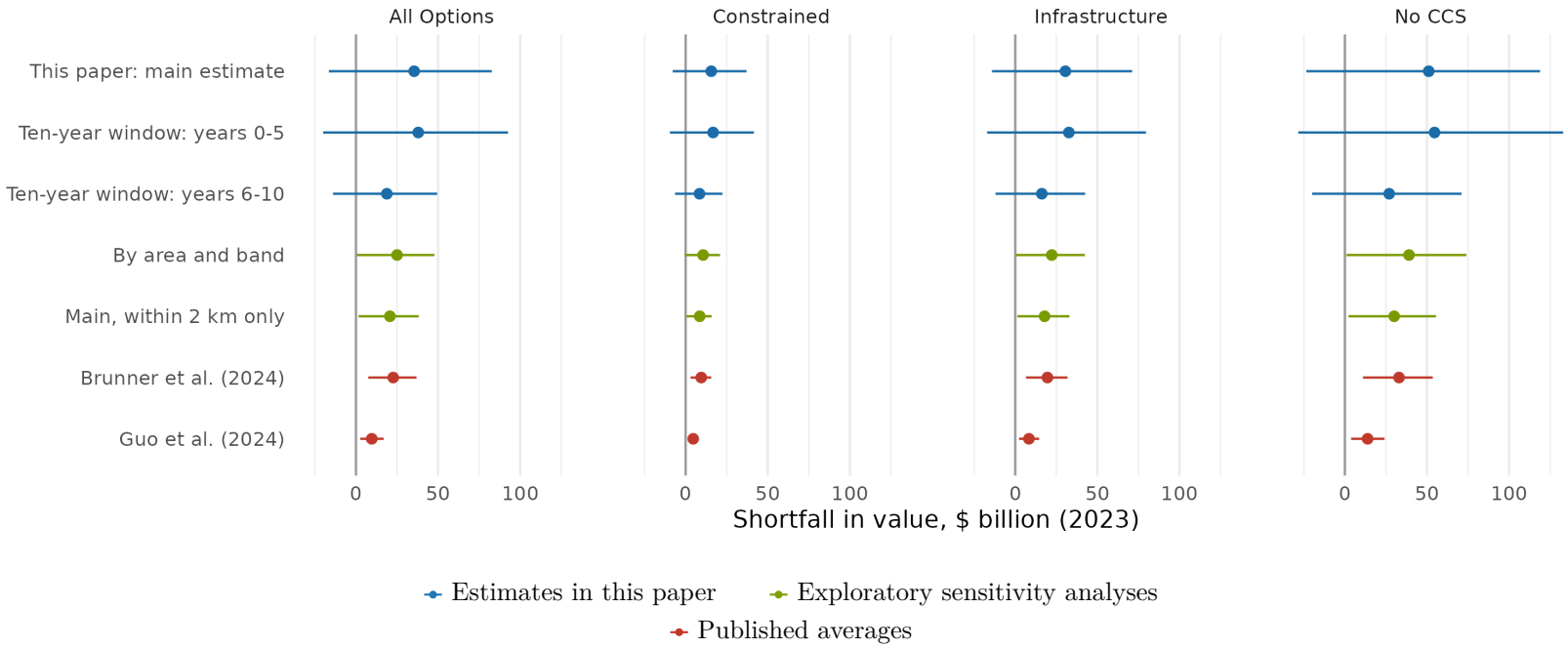
The price effects may fade over time. In the ten-year window, our estimates give $37.9 billion for years 0–5 and $18.8 billion for years 6–10 in All Options, and both intervals include

Table 8: Reduction in the value of newly exposed homes ($ billion, 2023)

	All Options	Constrained	Infrastructure	No CCS
Newly exposed parcels (thousands)	4431	2488	3872	6257
Their value, assessment-ratio-adjusted ($ billion)	970.3	450.2	833.7	1387.2
Assessed value adjusted by county / state / not adjusted (%)	97 / 3 / 0	95 / 5 / 0	96 / 4 / 0	97 / 3 / 0
A. Estimated in this paper ($ billion)				
Main estimate: years 0–5 after operation	35.4 [-16.5, 82.7]	15.6 [-7.9, 37.1]	30.5 [-14.2, 71.2]	51.0 [-23.5, 118.9]
Ten-year window (older developments): years 0–5	37.9 [-20.0, 92.5]	16.7 [-9.6, 41.6]	32.6 [-17.1, 79.6]	54.6 [-28.4, 132.9]
years 6–10	18.8 [-14.0, 49.4]	8.5 [-6.4, 22.4]	16.2 [-12.0, 42.5]	27.0 [-20.0, 71.0]
B. Published averages ($ billion)				
Brunner et al. (2024): average from announcement on	22.7 [7.5, 36.9]	9.5 [3.0, 15.6]	19.6 [6.6, 31.8]	33.0 [11.0, 53.5]
interval if their two bands were perfectly correlated	[1.6, 42.6]	[0.6, 18.0]	[1.5, 36.8]	[2.5, 61.8]
Guo et al. (2024): average after installation, in view within 5 km	9.6 [2.6, 16.8]	4.6 [1.3, 8.1]	8.3 [2.3, 14.6]	13.8 [3.8, 24.2]
C. Persistence assumptions for years 6–10 ($ billion)				
Full recovery (either study)	0	0	0	0
No recovery: Brunner et al. (2024)	22.7 [7.5, 36.9]	9.5 [3.0, 15.6]	19.6 [6.6, 31.8]	33.0 [11.0, 53.5]
No recovery: Guo et al. (2024)	9.6 [2.6, 16.8]	4.6 [1.3, 8.1]	8.3 [2.3, 14.6]	13.8 [3.8, 24.2]
D. Exploratory sensitivity analyses: years 0–5 after operation ($ billion)				
Main specification by area: rural and urban estimates in each band	25.0 [0.7, 47.8]	10.6 [-0.4, 21.0]	22.2 [0.5, 42.4]	39.0 [1.0, 73.9]
Main estimate within 2 km only, no effect beyond 2 km (assumed)	20.7 [1.5, 38.2]	8.6 [0.6, 15.9]	17.8 [1.3, 33.0]	30.0 [2.2, 55.5]
No effect	0	0	0	0

Notes: Reductions in the adjusted value of homes with no turbine within 10 km in 2023 and a new turbine within 5 km, in 2023 $ billion. Brackets show 95% intervals from 10,000 draws of the coefficients. Panel A uses the covariance of our two distance-band estimates. For Brunner et al., the first interval treats their two distance-band estimates as independent and the second as perfectly correlated. Panel D reports the two exploratory profiles, which we added after examining the main results. The intervals reflect sampling uncertainty only. Appendix table 12 reports the same calculations with unadjusted assessed values.

Figure 4: Reduction in the value of newly exposed homes by scenario and profile



Notes: Reductions in the adjusted value of homes within 5 km of a new turbine that had no turbine within 10 km in 2023, in 2023 $ billion, with 95% intervals. The exploratory profiles were added after examining the main results (table 8, panel D).

zero. For the published averages, the totals for years 6–10 are zero under full recovery and equal to the totals above under no recovery.

The two exploratory profiles give smaller totals than the main estimates. Applying separate rural and urban estimates gives $25.0 billion in All Options, because most exposed homes are rural, where the estimated decline is smaller. Applying the main estimate only within 2 km gives $20.7 billion. Both intervals exclude zero in All Options. Across all the profiles other than the no-effect benchmark, All Options totals range from $9.6 billion to $37.9 billion. This range reflects the choice of evidence and assumptions rather than a statistical interval.

The reductions fall mainly on rural homes. In the All Options scenario, rural tracts contain 85% of the newly exposed parcels and 81% of their adjusted value, and under our main estimates they account for $29.5 billion, or 83%, of the total. Parcels near more than one new development account for 70% of the newly exposed parcels and $21.9 billion, or 62%, of the total, with each parcel counted once. Illinois, North Carolina, Virginia, Indiana, and Pennsylvania together account for 46% of the total.

The calculation excludes homes that already have a turbine nearby. In the All Options scenario, a further 0.68 million parcels within 5 km of a new turbine already have an operating turbine within 10 km. Their adjusted value is $123 billion, and applying the profiles in panels A and B would give $1.3–5.0 billion. Finally, using unadjusted assessed values reduces the totals by 29–38%, which lowers the main All Options result to $22.4 billion (appendix table 12).

7 Discussion

Recent national studies appear to disagree about how much wind development reduces nearby property values. Our results suggest that much of this disagreement reflects the distances over which the effects are measured. Close to turbines, where almost every home can see one, distance and visibility capture nearly the same exposure, and our estimates are of similar magnitude to those of E. J. Brunner, Hoen, Rand, et al. (2024) and to the estimate of Guo et al. (2024) within 1.5 km. Averages that extend to homes several kilometers away are much smaller, which is why the published estimates give such different totals when applied to the scenarios.

The timing of the declines is informative about why prices fall. Much of the decline occurs before operation begins, and prices also fall around proposals that are never built. Buyers therefore appear to respond to proposals and the uncertainty surrounding them, and not only to operating turbines. Such uncertainty can fade once a project is built, and buyers who are less concerned about turbines may replace those who are more concerned, which is consistent with the smaller declines in later years found here and in previous studies. For rural homes, the decline begins with development and persists after operation, as expected if buyers respond to the project itself. The larger urban estimate instead rests on price movements several years before operation, which may reflect early responses to proposals or differences between local housing markets.

These differences matter for applying the historical estimates to future development. Future exposure is mostly rural, and the rural estimate is both more stable and drawn from a setting closer to the future one, so the lower total from separate rural and urban estimates may be a closer guide than our main total. Future developments would also be larger and often close to one another. Our estimates give no indication that larger developments produce larger declines, but the totals count each home once, so they would understate the reduction

if additional developments further reduce prices.

The case for compensating nearby owners depends on which estimates are closer to the truth. Under our main estimates, owners within 2 km of a new development would lose about $23,000 per home on average. If losses of this size are correct, they give nearby owners a strong reason to oppose construction, and governments may want to consider compensation to reduce this opposition, such as direct payments or subsidies to nearby owners or additional investment in host communities. Direct payments would match the losses more closely, since the declines are concentrated close to turbines, whereas the benefits of local investment, like those of tax revenues, are shared across the wider community (Hoen, E. Brunner, et al., 2025). Compensating all newly exposed owners in the All Options scenario, which adds 914 GW of wind capacity, would cost about $39,000 per MW under our main estimates and about $11,000 per MW under the Guo profile. If the smaller estimates are correct, compensation could be limited to the owners closest to turbines.

The design of compensation also depends on why prices fall. If buyers respond mainly to proposals, then payments that begin at operation would come too late for owners who sell during permitting and construction, and faster and more predictable siting decisions could shorten the period of lower prices. If they respond mainly to operating turbines, then payments tied to distance would better match the losses for as long as the effects last. The specific locations where turbines are built also matters. The Infrastructure scenario, which assumes improvements in transmission, permitting, and siting, adds 14% more wind capacity than All Options but would newly expose 13% fewer parcels, so policies that allow wind to be built farther from homes could reduce these losses for a given amount of capacity. Our calculations capture effects through the property market only and do not measure the overall welfare effects of wind development.

8 Conclusion

This paper combines historical evidence on property prices around US wind developments with an assessment of residential exposure under NREL's 2035 deployment scenarios. Prices within 2 km of new commercial developments fell by about 11% relative to prices 5–10 km away, and much of this decline appeared before operation began. The decline for rural homes was smaller, at about 6.2%, and less sensitive to the exclusion of individual developments. Together with recent national studies, our estimates suggest that price declines are concentrated within about 2 km of turbines and that buyers respond to proposals as well as to operating turbines.

The scenarios would place 2.5–6.3 million previously unexposed residential parcels within 5 km of a new turbine. Most are rural, a majority would be near several developments, and the developments would generally be larger than those in the historical sample. In the All Options scenario, applying our main estimates gives a conditional reduction of $35.4 billion in the value of these homes, and the estimates of Brunner et al. and Guo et al. give $22.7 billion and $9.6 billion, respectively. Across the nonzero profiles we report, totals in the Constrained scenario are 42–48% of those in All Options, while totals in the No CCS scenario are 44–56% higher. The reductions are modest relative to the value of the exposed homes, at 3.7% under our main estimates, but they fall mainly on rural owners and are largest per home for those closest to turbines.

Two questions remain important for future deployment. The first is how much of the price response comes from the proposal and development process rather than from operating turbines, which dates of project announcements would help to answer. The second is how prices respond around large developments and near several developments, settings that are common in the scenarios and will become easier to study as recent developments age.

More broadly, the effect of wind expansion on property values depends on where turbines are built, how close they come to homes, and how long the price effects last, as well as on the

amount of capacity added. If the larger estimates are correct, owners near new developments stand to lose enough to give them reason to oppose construction, and compensation through direct payments or additional local investment could reduce this opposition. Policies to expand wind power should therefore consider where capacity is sited and whether nearby owners are compensated, as well as how much capacity is built.

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Appendix A: Comparison With Previous Studies

Table 9 summarizes the recent US studies discussed in the introduction, including their samples, comparison groups, and the timing of their estimates.

Table 9: Recent US evidence on wind development and property values

Study	Data and sample	Comparison	Main estimate and timing
Hoen, Brown, et al. (2015)	CoreLogic and NY assessors; 50,000+ sales near 67 facilities	Before vs after, by distance	No statistically significant effect after announcement or construction
Dong et al. (2023)	Massachusetts and Rhode Island	Within 1 km vs 3–10 km	−2.5 to −4.6% after construction
E. J. Brunner, Hoen, Rand, et al. (2024)	CoreLogic 2005–2020; 428 projects of turbines >600 kW; 496,054 sales	0–1, 1–2, 2–3 miles vs 3–5 miles, by project	About −10.3% within 1 mile, through 10 years after announcement; about −2% and insignificant from about 5 years after operation
Guo et al. (2024)	ZTRAX 1997–2020; 5.7 million sales within 10 km	Turbine in view vs not, within 10 km	−1.1% on average; up to −8% within 1.5 km; peaks 3 years after installation, gone within about 10
Hoen, E. Brunner, et al. (2025)	CoreLogic (Cotality) 2005–2021; school districts	Districts with wind vs same-county districts without; preferred model excludes homes within 3.2 km	District-wide values about 3% higher, rising with project size

Appendix B: Construction of Synthetic Townships

Where official BLM cadastral data exist, we use real PLSS townships formed from the section boundaries in the BLM Ground Control Database (GCDB) First Division layer. We retain only standard sections (FRSTDIVTYP = "SN") with valid 15-character PLSSID codes and dissolve them into townships. Ohio is treated as fully synthetic because BLM coverage there is only around 19% of the state area, and including the sparse real coverage would produce a patchwork grid inconsistent with the rest of the state.

For all other areas without BLM coverage, we construct synthetic townships so that every location in the contiguous United States receives a township identifier. These areas include the original 13 colonies, Vermont, Maine, Kentucky, Tennessee, West Virginia, Texas, and Ohio, as well as Spanish and French land grants, tribal reservations, national parks, and miscellaneous digitization gaps. The gap area to be filled is the contiguous United States minus the buffered union of all real sections. A ± 100 m open and close buffer closes hairline gaps between adjacent real sections before the difference is taken, and fragments smaller than 1 km^2 are discarded as slivers.

Each gap polygon is assigned to a principal meridian (PM) using a Voronoi tessellation seeded from the 31 PM origins in the contiguous United States. Synthetic townships in a given area therefore inherit the coordinate frame of the historically closest PM, which makes their identifiers consistent with the real PLSS identifiers used in the same region. We then generate a synthetic township grid for each PM zone using the same geometric logic as the real survey. Townships are 6 miles square, with longitude widths computed at the latitude of the correction-band baseline to account for meridian convergence, and correction bands reset the reference meridian every four townships, as in the standard PLSS surveying convention.

Synthetic townships are clipped at state lines, matching the practice of the original PLSS surveyors, who stopped lines at jurisdictional boundaries rather than extending them across.

A synthetic grid cell that straddles a state line therefore produces two separate polygons with distinct identifiers. For example, a cell on the Kentucky–Tennessee border yields one KY-5PM-T*n*N-R*r*W township and one TN-5PM-T*n*N-R*r*W township, so no synthetic township crosses a state line and every township identifier maps to a single state.

Township identifiers take the form {state}-{PM}-T{n}{dir}-R{r}{dir}, as in IA-5PM-T3N-R7W. A source flag distinguishes real from synthetic townships, and an is_partial flag marks any synthetic township whose clipped area is below 95% of the area of its full grid cell. All area calculations use the CONUS Albers Equal Area projection (EPSG 5070), and geometries are stored in WGS84 (EPSG 4326). We assign each property location to the township that contains it, and a location on a shared boundary is assigned to the township with the smallest identifier.

Appendix C: Implicit Regression Weights

The Frisch–Waugh–Lovell theorem expresses a coefficient or phase contrast as a weighted sum of outcomes. Let $\widetilde{X}$ be the design matrix, including event-time interactions and controls, after weighted removal of fixed effects. Let W be the diagonal matrix of estimation weights and L the vector defining the contrast, with zero entries for the control coefficients. The observation weights are

$$\omega = W\widetilde{X}(\widetilde{X}'W\widetilde{X})^{-1}L,$$

so the estimate is $\sum_i \omega_i \ln p_i$. If outcomes consist of fixed effects, controls, and a treatment effect τ_i on each treated sale, its treatment-effect component is $\sum_i \omega_i \tau_i$ over treated observations.

We aggregate the weights on treated sales by development and event year and compare them with the target weights of Wing et al. (2024), which are based on each development's share of treated sales. The reported distance is half the sum of absolute differences between the two sets of weights over the development and operating years, excluding the baseline. For a years 0–5 contrast, the weights on treated sales sum to one over those years and to minus one over the baseline years, so negative baseline weights are part of the intended contrast. Across treated and comparison observations together, the weights sum to zero within each development-year because of the corresponding fixed effects.

The main model within 2 km has a weight distance of 0.13 at the development-year level and at the level of individual sales. Weights on treated sales in event years outside the baseline and years 0–5 sum to 0.06 in absolute value. The negative development-year weights on treated sales in years 0–5 sum to less than 0.001 in absolute value in each of the four models. The development with the largest weight has 12% of the weight on years 0–5, compared with a maximum of 9% under the target weights. The corresponding distance is

0.14 for rural homes, 0.11 for urban homes, and 0.10 for the 2–5 km model. The largest rural and urban developments carry 8% and 22%, respectively, against targets of 7% and 18%.

We summarize weight concentration by the inverse sum of squared development shares of the years 0–5 weight. It is equivalent to about 20 equally weighted developments for the main estimate, 37 for rural homes, and 7 for urban homes. This measures concentration rather than the number of developments or independent clusters. The diagnostics show little negative weighting but substantial concentration in the main and urban samples.

Appendix D: Assessment Valuation Diagnostics

Table 10 compares 2023 CoreLogic calculated total values with the HPI-adjusted prices of the latest arm's-length residential sales in 2021–2023. Following International Association of Assessing Officers (2013), it reports median assessment-to-price ratios, coefficients of dispersion, and price-related differentials.

In most states, the median assessment-to-price ratio lies between 0.7 and 0.9. It is lower in several states with fractional or older-base assessments, including Delaware (0.15), Illinois (0.28), New York (0.39), Michigan (0.40), and Pennsylvania (0.47). Dispersion is also substantial in some states, with coefficients of dispersion of 113.2 in New York, 69.8 in Missouri, and 54.5 in Pennsylvania, compared with the standard's residential ranges of at most 15, or 20 in rural areas (International Association of Assessing Officers, 2013).

Our processed deed panel covers the states with turbines. For Alabama, the District of Columbia, Florida, Georgia, Louisiana, and South Carolina, we instead read the raw 2021–2023 deeds using the same transaction filters and take residential status from the 2023 snapshot. Their median ratios are 0.81–0.91, except in Louisiana, where the ratio is 0.077.

In Louisiana, CoreLogic's calculated value comes from the assessed-value field. Its median ratio to the assessor's separate market-value field is 0.10, consistent with the state's assessment fraction for land and residential improvements (La. Const. art. VII, §18(B)). The comparison with sale prices gives an interquartile range of 0.07–0.09 and a coefficient of dispersion of 20.

The Louisiana adjustment factor of about 13 comes from the empirical ratio of assessments to sale prices, and we apply no statutory factor in addition. It raises the All Options value of the state's newly exposed parcels from $0.6 billion to $7.5 billion. The median ratio to the assessor's market value and the median ratio to sale prices come from different comparisons, so dividing one by the other would not give the typical ratio of assessor market value to sale price.

The county and state adjustments account for differences in assessment levels between areas but not within counties. In New York, for example, municipalities within a county can assess at different fractions of market value.

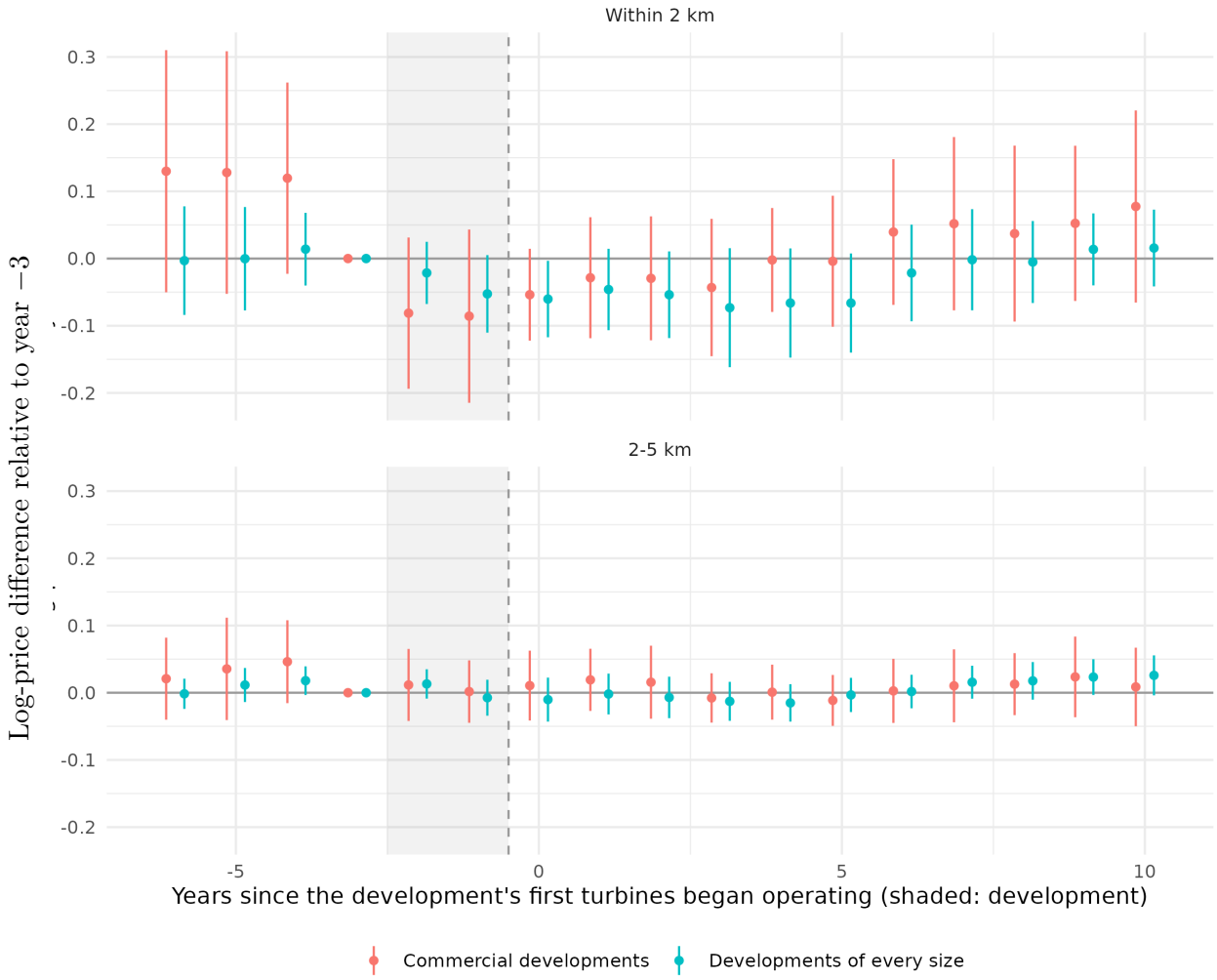
Table 10: Assessment-to-price ratios, 2021–2023

State	Sales	Median ratio	Interquartile range	COD	PRD
AL†	177,376	0.90	0.77–1.03	24.7	1.08
AR	133,768	0.74	0.62–0.90	29.7	1.08
AZ	404,742	0.76	0.65–0.85	17.8	1.00
CA	1,012,824	0.85	0.83–0.95	10.4	1.01
CO	322,419	0.92	0.81–0.99	13.1	1.02
CT	134,948	0.52	0.45–0.60	18.9	1.03
DC†	22,973	0.84	0.78–0.94	12.8	1.04
DE	47,797	0.15	0.11–0.20	40.8	1.17
FL†	1,441,312	0.81	0.74–0.90	13.7	1.02
GA†	522,754	0.91	0.81–0.96	13.1	1.02
IA	152,816	0.91	0.83–1.00	14.9	1.04
ID	75,723	0.85	0.72–1.04	30.6	1.07
IL	232,470	0.28	0.26–0.31	14.9	1.03
IN	266,848	0.84	0.72–0.95	21.4	1.05
KS	75,900	0.87	0.77–0.97	18.1	1.02
KY	157,968	0.83	0.82–0.95	12.2	1.01
LA†	158,253	0.08	0.07–0.09	20.0	1.04
MA	235,747	0.79	0.73–0.86	11.6	1.02
MD	253,512	0.78	0.68–0.86	15.4	1.02
ME	39,853	0.70	0.54–0.89	39.6	1.11
MI	401,351	0.40	0.36–0.46	19.0	1.03
MN	174,946	0.89	0.82–0.97	11.3	1.02
MO	236,456	0.53	0.16–0.82	69.8	1.12
MS	56,699	0.56	0.45–0.71	35.7	1.11
MT	33,408	0.87	0.75–1.04	26.8	1.06
NC	507,997	0.73	0.59–0.89	26.6	1.05
ND	22,055	0.76	0.47–0.90	36.5	1.13
NE	88,847	0.82	0.73–0.90	15.9	1.03
NH	52,641	0.78	0.65–0.92	25.1	1.06
NJ	341,719	0.62	0.50–0.77	28.9	1.02
NM	58,318	0.72	0.53–0.86	33.4	1.02
NV	186,634	0.61	0.46–0.77	30.6	1.03
NY	494,978	0.39	0.06–0.75	113.2	1.70
OH	430,571	0.80	0.68–0.91	21.3	1.04
OK	170,248	0.88	0.81–0.95	15.1	1.01
OR	180,433	0.91	0.82–1.02	15.2	1.02
PA	486,499	0.47	0.29–0.65	54.5	1.14
RI	35,993	0.77	0.67–0.86	16.3	1.05
SC†	223,475	0.82	0.75–0.94	16.5	1.02
SD	7,390	0.80	0.69–0.92	22.9	1.06
TN	370,506	0.69	0.58–0.82	24.1	1.06
TX	965,273	0.92	0.80–1.05	20.7	1.03
UT	120,021	0.86	0.75–0.99	21.1	1.03
VA	369,910	0.86	0.78–0.93	12.3	1.01
VT	36,220	0.67	0.55–0.86	40.3	1.20
WA	278,995	0.84	0.76–0.93	13.7	1.04
WI	218,191	0.80	0.64–0.94	25.5	1.06
WV	45,995	0.45	0.38–0.53	26.3	1.06
WY	16,248	0.88	0.76–1.04	26.9	1.03
All	12,482,020	0.80	0.62–0.92	26.9	1.03

Notes: The numerator is the 2023 CoreLogic calculated total value, and the denominator is the HPI-adjusted price of the latest arm's-length residential sale in 2021–2023. Conflicting prices on that date are excluded, as are log ratios more than three interquartile ranges beyond state quartiles. COD: coefficient of dispersion (%). PRD: price-related differential. † Ratios use raw deeds because no processed sales panel exists, with the same transaction filters and 2023 residential status.

Appendix E: Additional Results

Figure 5: Persistence of price differences over ten years



Notes: Developments that began operating by 2014. Points show annual log-price coefficients relative to year −3, with 95% confidence intervals. Comparison homes are 5–10 km from the same development, and shaded years are the development phase.

Table 11: Sensitivity to influential developments

Excluded	Turbines	Year	Weight (%)	Treated sales	Years 0–5 [95% interval]
A. Main estimate: all homes					
Nothing (the reported estimate)			0	34,366	−0.116 [−0.224, −0.007]
Testa Produce, IL	1	2011	12.3	30,040	−0.066 [−0.148, 0.016]
Lake Palmdale, CA	1	2004	9.8	31,136	−0.139 [−0.244, −0.034]
Lincoln Electric, OH	1	2011	9.8	31,231	−0.089 [−0.201, 0.022]
California State Prison Los Angeles County, CA	1	2017	6.0	32,412	−0.116 [−0.230, −0.001]
Equity Industrial Partners, MA	3	2012	5.6	32,509	−0.118 [−0.232, −0.005]
All five			43.4	19,864	−0.035 [−0.075, 0.005]
Developments in Massachusetts			22.3	26,932	−0.139 [−0.270, −0.008]
Developments in California			22.0	26,605	−0.156 [−0.268, −0.043]
Developments in Ohio			16.7	28,897	−0.097 [−0.215, 0.021]
B. Rural homes					
Nothing (the reported estimate)			0	14,392	−0.064 [−0.105, −0.023]
Kirkwood Community College, IA	1	2012	8.2	13,333	−0.071 [−0.111, −0.031]
Mount Wachusett Wind Farm, MA	2	2009	6.3	13,603	−0.065 [−0.108, −0.022]
Lakefield, MN	137	2011	4.9	13,694	−0.056 [−0.093, −0.019]
Kenston School District, OH	1	2012	4.1	13,829	−0.068 [−0.110, −0.026]
Whirlpool Corporation - Greenville Wind Farm, OH	3	2018	3.8	13,879	−0.061 [−0.102, −0.020]
All five			27.3	10,770	−0.058 [−0.098, −0.018]
Developments in Massachusetts			20.8	11,626	−0.065 [−0.115, −0.016]
Developments in Iowa			14.1	12,426	−0.077 [−0.119, −0.035]
Developments in Ohio			13.2	12,558	−0.073 [−0.117, −0.029]

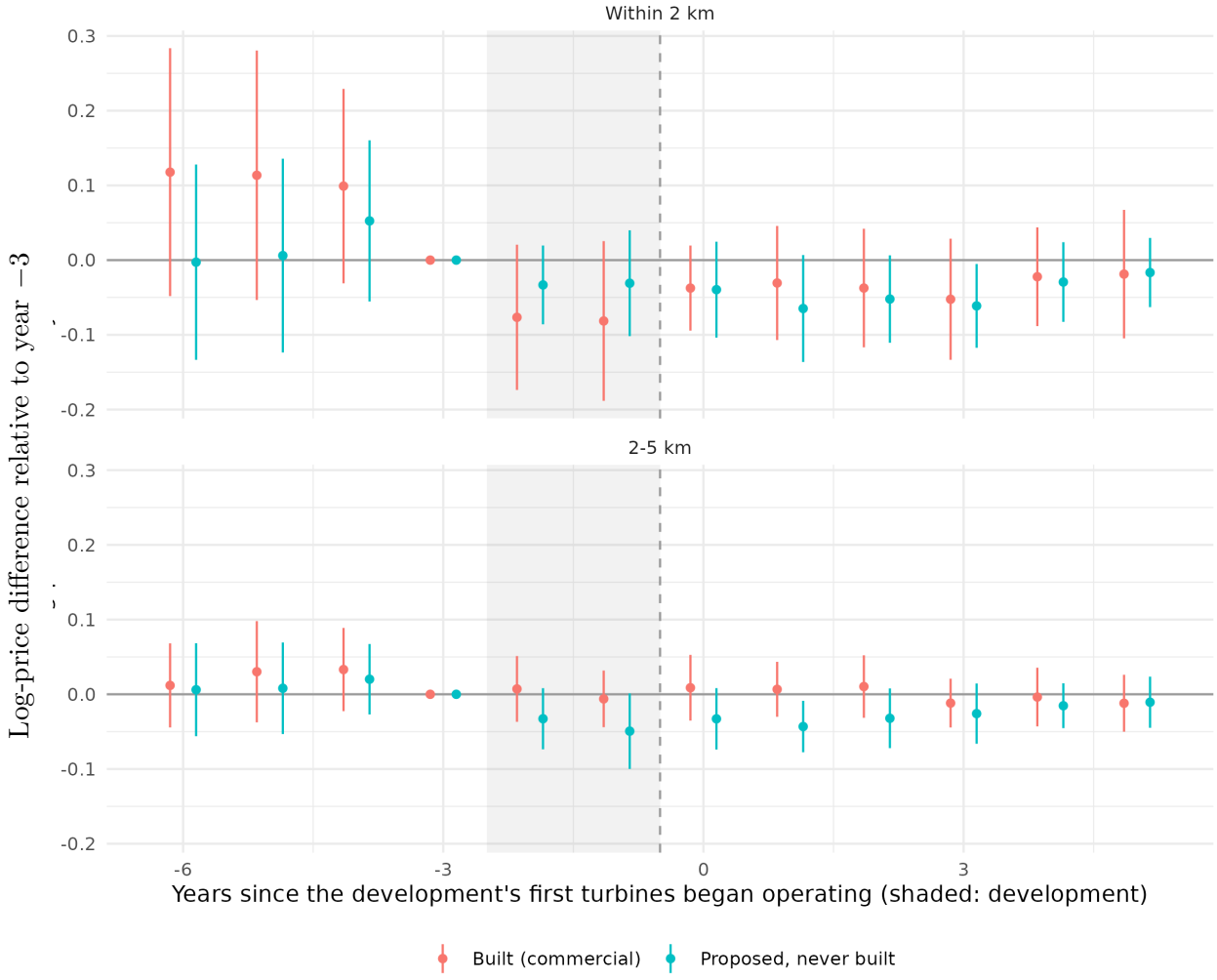
Notes: Years 0–5 contrasts from a regression of log sale price, with county-clustered 95% intervals. The model is re-estimated after excluding each of the five developments with the largest weights, all five together, and the developments assigned to each of the three states with the largest weights. A development's observations are removed regardless of the homes' states. Samples and corrective weights are rebuilt for every refit. Weight is the excluded share of the original implicit weight on years 0–5. Names follow the US Wind Turbine Database.

Table 12: Reduction in the value of newly exposed homes using unadjusted assessed values ($ billion, 2023)

	All Options	Constrained	Infrastructure	No CCS
Newly exposed parcels (thousands)	4431	2488	3872	6257
Their value, assessed ($ billion)	611.1	311.2	530.9	895.0
Assessed value adjusted by county / state / not adjusted (%)	97 / 3 / 0	95 / 5 / 0	96 / 4 / 0	97 / 3 / 0
A. Estimated in this paper ($ billion)				
Main estimate: years 0–5 after operation	22.4 [-10.4, 52.2]	10.9 [-5.4, 25.7]	19.5 [-9.0, 45.5]	33.0 [-15.1, 76.8]
Ten-year window (older developments): years 0–5	23.9 [-12.6, 58.4]	11.6 [-6.6, 28.9]	20.9 [-10.9, 50.9]	35.3 [-18.3, 85.8]
years 6–10	11.9 [-8.8, 31.2]	5.9 [-4.4, 15.5]	10.3 [-7.6, 27.2]	17.4 [-12.9, 45.9]
B. Published averages ($ billion)				
Brunner et al. (2024): average from announcement on	14.3 [4.8, 23.2]	6.6 [2.1, 10.9]	12.5 [4.2, 20.3]	21.3 [7.2, 34.5]
interval if their two bands were perfectly correlated	[1.1, 26.8]	[0.4, 12.5]	[1.0, 23.4]	[1.7, 39.8]
Guo et al. (2024): average after installation, in view within 5 km	6.0 [1.7, 10.5]	3.2 [0.9, 5.6]	5.3 [1.5, 9.2]	8.9 [2.4, 15.5]
C. Persistence assumptions for years 6–10 ($ billion)				
Full recovery (either study)	0	0	0	0
No recovery: Brunner et al. (2024)	14.3 [4.8, 23.2]	6.6 [2.1, 10.9]	12.5 [4.2, 20.3]	21.3 [7.2, 34.5]
No recovery: Guo et al. (2024)	6.0 [1.7, 10.5]	3.2 [0.9, 5.6]	5.3 [1.5, 9.2]	8.9 [2.4, 15.5]
D. Exploratory sensitivity analyses: years 0–5 after operation ($ billion)				
Main specification by area: rural and urban estimates in each band	15.5 [0.4, 29.6]	7.5 [-0.2, 14.7]	14.4 [0.4, 27.4]	25.2 [0.7, 47.7]
Main estimate within 2 km only, no effect beyond 2 km (assumed)	13.1 [1.0, 24.2]	6.0 [0.4, 11.1]	11.5 [0.9, 21.3]	19.4 [1.4, 35.9]
No effect	0	0	0	0

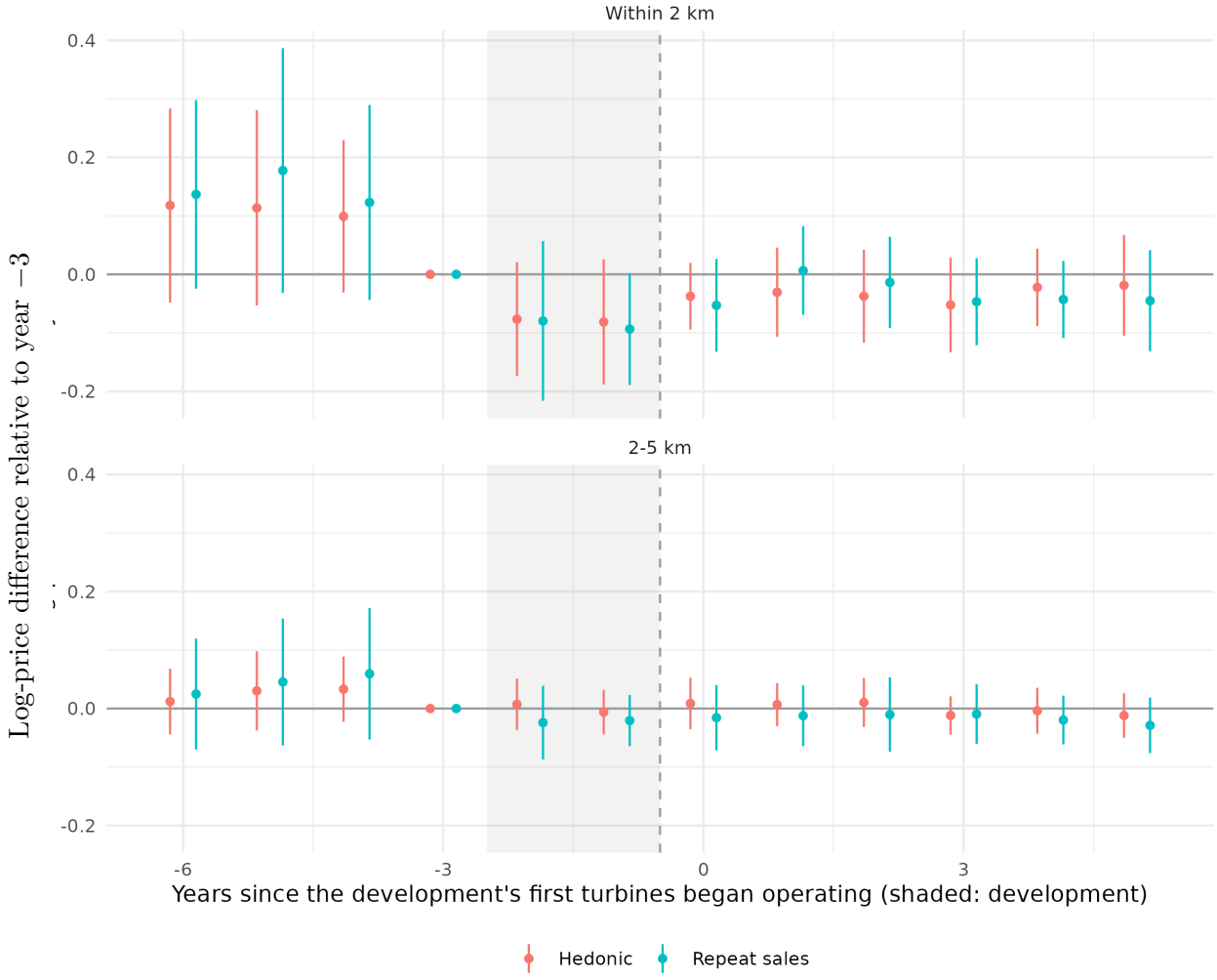
Notes: As table 8, but using CoreLogic's calculated total values without the assessment-ratio adjustment.

Figure 6: Built developments and proposals never built



Notes: Commercial developments, five-year follow-up. Points show annual log-price coefficients relative to year −3, with 95% confidence intervals. Comparison homes are 5–10 km from the same development, and shaded years are the development phase.

Figure 7: Hedonic and repeat-sales estimates



Notes: Commercial developments, five-year follow-up. Points show annual log-price coefficients relative to year −3, with 95% confidence intervals. Comparison homes are 5–10 km from the same development, and shaded years are the development phase.